Emergent Spacetime and the Dark Sector

Probing Dark Matter via the ER = EPR Conjecture in Holographic Tensor Networks

CANDIDATE	SUPERVISOR	ACADEMIC SCOPE	SUBMISSION DATE
[Candidate Name]	[Advisor / Group Head]	40 Weeks • Full Master's Thesis	October 2026

EXECUTIVE ABSTRACT

The missing dynamical mass observed in galactic rotation curves and large-scale cosmic structure remains one of the central puzzles in modern physics. While the standard ΛCDM paradigm posits non-baryonic elementary particles, decades of cryogenic direct detection and accelerator searches have yielded null bounds. In this thesis, we propose an alternative framework grounded in the holographic principle and the ER = EPR conjecture, modeling dark matter not as an ad-hoc particle field, but as an emergent gravitational manifestation of non-local quantum entanglement defects within the vacuum geometry. Treating continuous spatial slices as discrete quantum tensor networks, we introduce ensembles of non-local entanglement bonds—Planckian Einstein–Rosen bridges lacking Standard Model gauge charges—superimposed onto the background metric. Using the discrete Ryu–Takayanagi minimal cut formulation and the first law of entanglement thermodynamics, we derive the effective stress-energy backreaction $\langle T_{\mu\nu}^{\text{eff}} \rangle$ in the weak-field limit, benchmark against empirical SPARC galaxy datasets, and test whether flat rotation curves emerge directly from quantum information geometry.

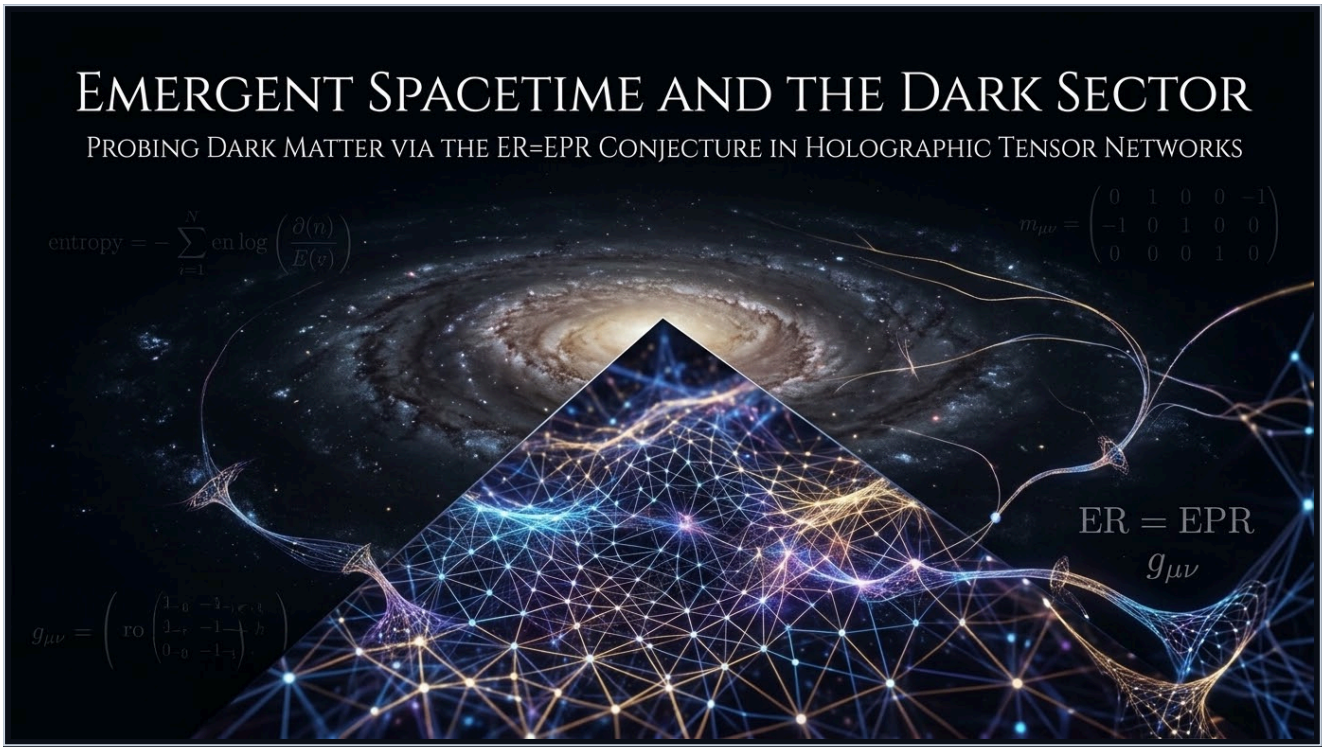


Figure 1: Multi-Scale Geometric Transition. Macroscopic galactic rotation (top) anchored to its microscopic holographic entanglement substrate (bottom). Microscopic Planckian ER bridges link distant vacuum sites, generating geometric backreaction without Standard Model matter fields.

1. THEORETICAL FORMALISM & ENTANGLEMENT BACKREACTION

In holographic duality, spatial geometry is an emergent property of boundary quantum states governed by the Ryu–Takayanagi formula, relating bulk geodesics to entanglement entropy: $S(A) = Area(\gamma_A) / (4G\hbar)$. We treat cold dark matter as topological entanglement defects without localized gauge charges.

A. FIRST LAW OF ENTANGLEMENT

Perturbations in the density matrix satisfy $\delta\langle H_{mod}\rangle = \delta S_{ent} / (2\pi)$. In semiclassical gravity, this variation couples directly to local Riemannian curvature, generating an effective gravitational backreaction $\delta R_{\mu\nu}$ mimicking a distributed dust halo.

B. NON-LOCAL ER BRIDGE ENSEMBLES

Microscopic wormholes lack open string endpoints, giving them zero color, weak, or electromagnetic charge. Their physical interaction is strictly confined to geometric metric distortion, satisfying cold collisionless dark matter constraints.

$$v_{circ}^2(r) = r \cdot (\partial\Phi / \partial r) \approx (r\,c^2 / 2) \cdot (\partial g_{00}^{eff} / \partial r) \sim const \quad \Rightarrow \quad Asymptotic\ Flatness$$

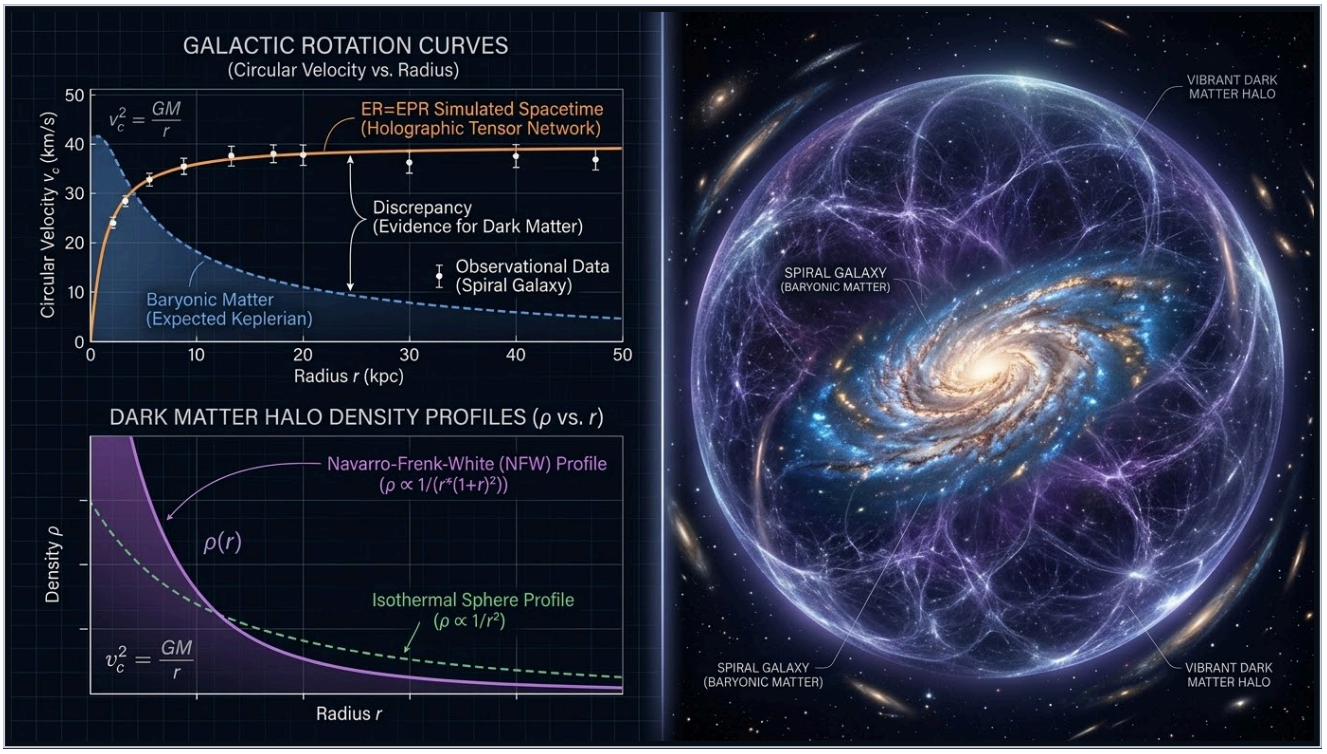


Figure 2: Empirical Target Profiles & Galactic Lensing Halo. (Left) Observed flat galactic rotation curve contrasted against Keplerian Newtonian decline ($v_c \propto r^{-1/2}$) alongside NFW and isothermal profiles. (Right) Artistic rendering of the extended halo envelope surrounding the baryonic disk.

2. COMPUTATIONAL PIPELINE & ALGORITHMIC ARCHITECTURE

To maintain empirical falsifiability, the theoretical model is implemented as a discrete numerical simulation engine using tensor networks and graph-theoretic min-cut solvers across 2D spatial slices.

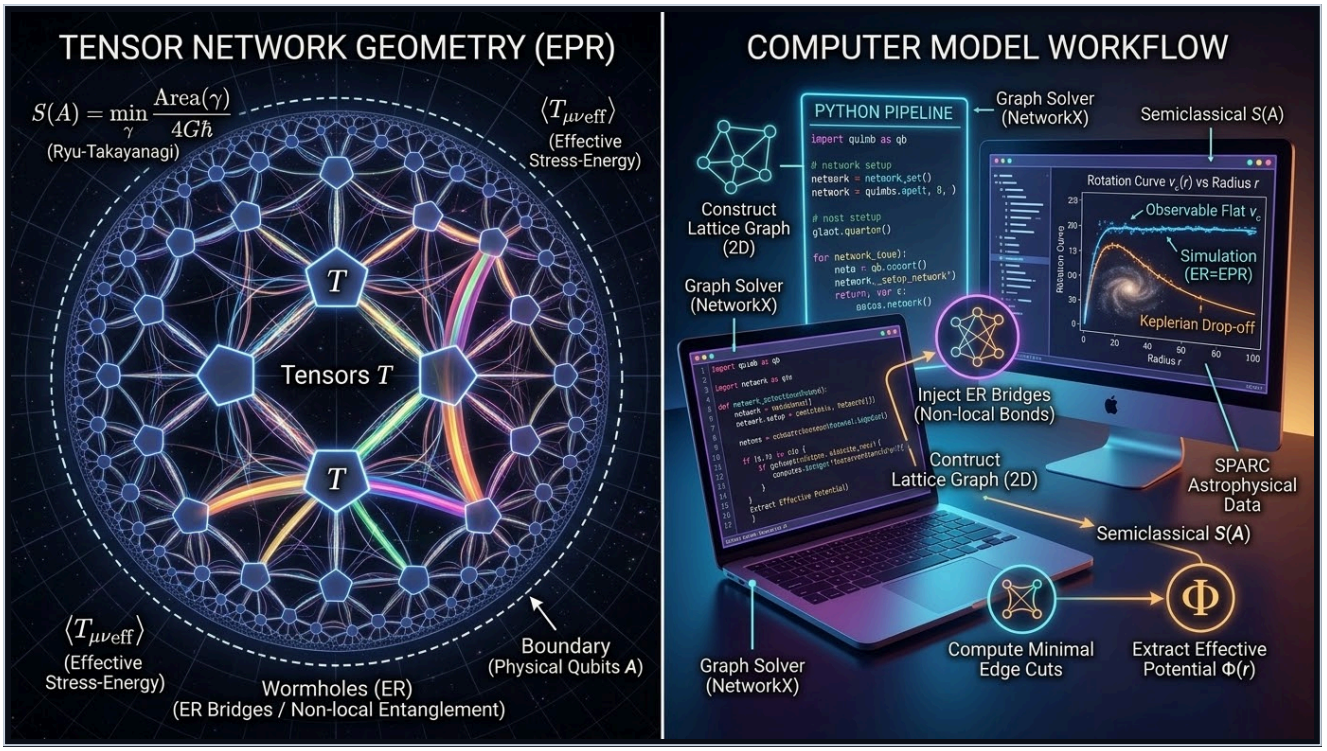


Figure 3: Discrete Tensor Network Implementation & Python Workflow. (Left) 2D hyperbolic tensor network discretized on a Poincaré disk with injected non-local ER bridges. (Right) Complete Python pipeline (Quimb, NetworkX) computing min-cuts and extracting rotation curves against SPARC galaxy data.

3. ALGORITHMIC MODULES & TECHNICAL SPECIFICATIONS

Pipeline Stage	Algorithmic Implementation	Deliverable
1. Lattice Generator	Discrete hyperbolic tessellation using MERA and HaPPY pentagon tensor networks.	Baseline spatial vacuum metric $g_{\mu\nu}^{0}$.
2. Defect Injection	Non-local bond insertions scaled by bond dimension χ to simulate ER bridge density.	Perturbed adjacency & entropy graph.
3. RT Min-Cut Solver	Push-Relabel and Boykov–Kolmogorov algorithms calculating minimal edge cuts.	Entanglement area profile *Area(γ_A)*.
4. Metric Inversion	Numerical differentiation of minimal surfaces to extract gravitational potential $\Phi_{\text{eff}}(r)$.	Radial circular velocity profiles $v_c(r)$.
5. Astrophysical Fitting	Non-linear regression against empirical rotation data from the SPARC database.	χ^2-goodness of fit and residual metrics.

4. 40-WEEK MASTER TIMELINE & KEY MILESTONES

Phase	Timeline	Core Research Tasks	Formal Milestone
Phase 1	Weeks 1–10	Literature review; derive $\langle T_{\mu\nu}^{\text{eff}} \rangle$; verify w = 0 dust equation of state.	Draft Ch. 1 & 2; analytical metric derivation.
Phase 2	Weeks 11–22	Build 2D hyperbolic lattice; implement RT min-cut solver; benchmark flat/AdS geodesics.	Validated Python repo; Ch. 3 draft.
Phase 3	Weeks 23–32	Inject non-local ER bridges; extract potential Φ(r); fit curves against SPARC database.	Final simulation plots; Ch. 4 & 5 drafts.
Phase 4	Weeks 33–40	Synthesize results; supervisor reviews; prepare defense presentation and archive codebase.	Complete Thesis Submission & Defense.

5. COMPLETE SEVEN-CHAPTER DISSERTATION MASTER SYLLABUS

Chapter 1: The Missing Mass Dilemma & Galactic Dynamics **Weeks 1–5**

Derivation of Keplerian velocity profiles; observational flat rotation curves; ΛCDM particle candidates; null experimental direct detection bounds; MOND and modified gravity tensions.

Chapter 2: The Quantum Geometry of Spacetime (ER = EPR) **Weeks 6–10**

Entanglement as the microscopic glue of geometry; Einstein–Rosen bridges; EPR pairs; the Maldacena–Susskind conjecture; Planckian wormholes without gauge charges.

Chapter 3: Holographic Entanglement & Tensor Networks **Weeks 11–16**

AdS/CFT correspondence; Ryu–Takayanagi formula; First Law of Entanglement; Faulkner et al. derivation of Einstein equations; discrete MERA and HaPPY codes.

Chapter 4: The Entanglement Defect Model of Dark Matter **Weeks 17–22**

Analytical formulation of non-local ER bridge ensembles; derivation of the effective stress-energy tensor $\langle T_{\mu\nu}^{\text{eff}} \rangle$; proof of pressureless dust behavior ($w \approx 0$); weak-field metric potential.

Chapter 5: Discrete Numerical Engine & Algorithmic Architecture **Weeks 23–27**

Discretization scheme; hyperbolic graph generation in Quimb and NetworkX; Ford–Fulkerson min-cut algorithms; central baryonic mass injection; metric inversion routines.

Chapter 6: Simulation Runs, Galaxy Fitting & SPARC Benchmarks **Weeks 28–34**

Numerical baseline verification; emergence of flat velocity profiles; non-linear fits against SPARC galaxy data (NGC 3198, NGC 2841); comparison with NFW and MOND profiles.

Chapter 7: Discussion, Astrophysical Implications & Outlook **Weeks 35–40**

Physical interpretation of dark matter as entanglement defect geometry; equivalence principle consistency; extension to 3+1D and cosmic lensing; thesis summary and open questions.